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Effect of Regioisomerism on the Efficiency of 1-Phenylpyrrole-Type Atropisomeric Amino Alcohol Ligands in Enantioselective Organometallic Reactions

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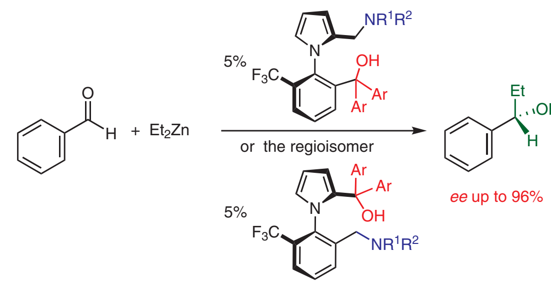

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Abstract Syntheses of two regioisomeric series of atropisomeric amino alcohols and a comparative study on their application in the enantioselective addition of diethylzinc to benzaldehyde are reported. Systematic modification of the electronic and steric properties of the functional groups resulted in highly efficient catalyst ligands in both series. Quantum-chemical calculations agreed well with the experimental results of this first systematic comparative study on regioisomeric atropisomeric ligands.

Key words atropisomerism, amino alcohols, enantioselectivity, asymmetric catalysis, phenylpyrrole, ligands

In the last 40 years, optically active amino alcohols have become more and more important as ligands for various enantioselective catalytic reactions.[1–4] Beside asymmetric transfer hydrogenation[5,6] and hydroamination,[7] several asymmetric iodinations,[8] enantio- and/or diastereoselective alkynylations and arylations,[9–12] asymmetric imino-Reformatsky reactions,[13,14] 1,3-dipolar cycloadditions,[15] and numerous variations on enantioselective alkylation of prochiral aldehydes[16–20] have been developed by using chiral amino alcohols as ligands. In the most cases, the ligands used contained one or two asymmetric carbon atoms with the amino and hydroxy functions situated in the α- and β-positions. Only a few atropisomeric amino alcohols have been reported in the literature (compounds **1**–**5** in Figure 1).[8,19–28]

Moreover, the effects of rigid and bulky structural units of a few ligands on chiral induction have been discussed in several cases,[7,16,17] and the crystal structure of a chiral amino alcohol and that of its tantalum complex have also been reported.[24] However, systematic investigations of regioisomerism in the case of atropisomeric ligands are absent from the literature. We have recently reported successful applications of the first representatives of 1-phenylpyrrole-derived atropisomeric amino alcohols (Scheme 1; compounds **6** and **7**).[20,25–28] The results of the enantioselective addition of diethylzinc to aromatic aldehydes demonstrated significant differences between the asymmetric-induction

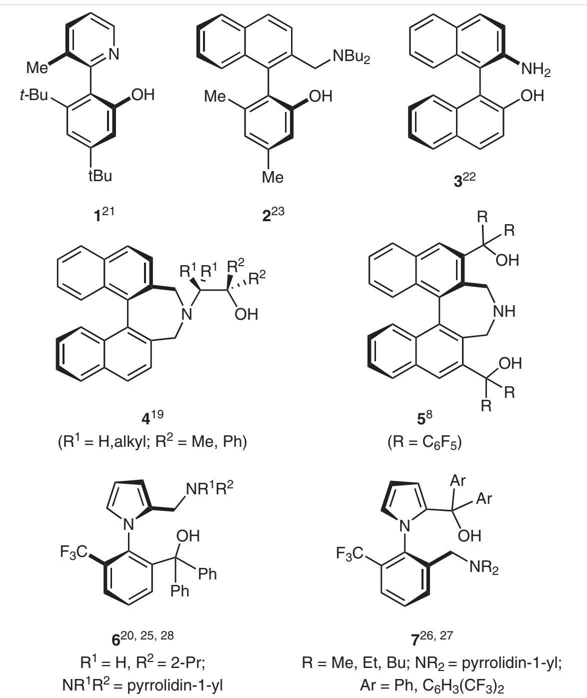
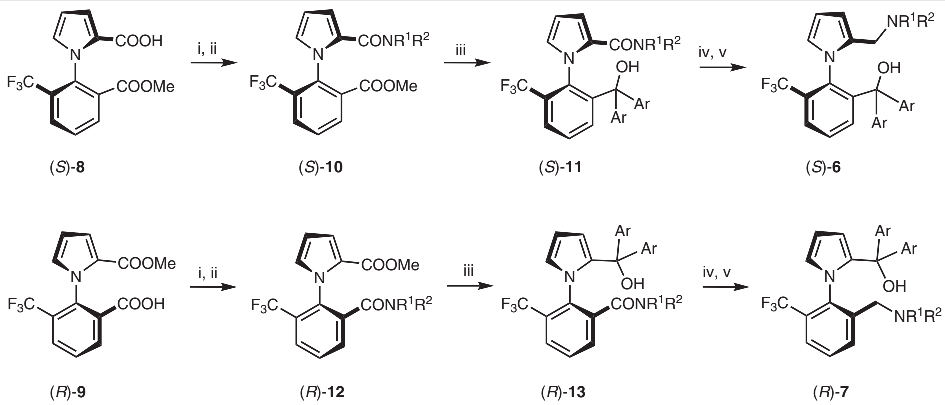


Figure 1 Atropisomeric amino alcohol ligands



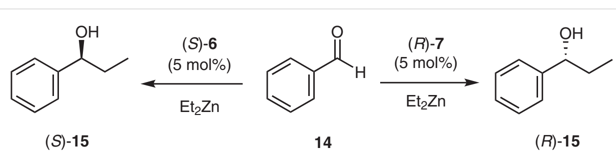
6 or 7	NR^1R^2	Ar
a	NMe_2	Ph
b	NEt_2	Ph
c	NBu_2	Ph
d	pyrrolidin-1-yl	Ph
e	(S)-N(Me)$(CH_2)_2$Ph	Ph
f	(S)-N(Me)$(CH_2)_2$Ph	Ph
g	NEt_2	3-$(F_3C)C_6H_4$
h	NEt_2	3,5-$(F_3C)_2C_6H_3$
i	NMe_2	3-$(F_3C)C_6H_4$
j	pyrrolidin-1-yl	3-$(F_3C)C_6H_4$
k	morpholin-4-yl	3-$(F_3C)C_6H_4$
l	NMe_2	3,5-$(F_3C)_2C_6H_3$
m	pyrrolidin-1-yl	3,5-$(F_3C)_2C_6H_3$

Scheme 1 Syntheses of (*S*)-**6a–h**, (*R*)-**7a–d**, and (*R*)-**7i–m**. *Reaction conditions*: (i) $SOCl_2$, toluene; (ii) HNR^1R^2; (iii): ArMgBr, Et_2O; (iv) $BH_3{\cdot}SMe_2$, toluene; (v) NaOH, MeOH.

properties of the two regioisomeric ligands (Series **6a–k** and Series **7a–d**, and **7i–m**). To shed light on the role of the electronic and steric properties of the substituents connected to the amino-nitrogen-substituted and the hydroxy-group-substituted carbon atom, we decided to prepare several new atropisomeric ligands type **6** and **7**. Furthermore, we report the first comparative study on the consequences of interchange of the positions of the dialkylamino and the hydroxymethyl group on the 1-(2-trifluoromethyl)phenylpyrrole skeleton (regioisomerism).

Both series of amino alcohols [(*S*)-**6a–k** and (*R*)-**7a–k**] were prepared starting from the regioisomeric optically active 1-[2-(methoxycarbonyl)-6-(trifluoromethyl)phenyl]-1*H*-pyrrole-2-carboxylic acids[29] (S_a)-**8**[25] and (R_a)-**9**[26] (Scheme 1). The reaction steps of both pathways were optimized separately, and the enantiomeric purities of the intermediates were checked by HPLC; the products were shown to be enantiomerically pure (ee >99%). The atropisomeric intermediates and the products were stereochemically stable under the investigated conditions. Experimental details are described in the Supporting Information.

The asymmetric-induction properties of the two regioisomeric amino alcohol series were tested in the enantioselective addition of diethylzinc to benzaldehyde in hexane (Scheme 2).



Scheme 2 Enantioselective addition of diethylzinc to benzaldehyde (**14**) in the presence of ligands (*S*)-**6** or (*R*)-**7**.

Each reaction was carried out in the presence of 5 mol% of (*S*)-**6a–h**, (*R*)-**7a–d**, or (*R*)-**7i–m** with a 1 M solution of diethylzinc in hexane.[30] Full conversion was achieved in all cases, and the products (*S*)-**15** or (*R*)-**15** were isolated in excellent yields of 85–96%, except for the reaction in the presence of (*S*)-**6f**, which gave a 30% yield. The ee values of the alcohol products could be used to indicate the asymmetric-induction capabilities of the ligands, and the results are collected in Table 1. Temperature-dependent measurements showed that ligands (*S*)-**6a–h** gave their best results at 0 °C, whereas the regioisomeric ligands (*R*)-**7a–d** and (*R*)-**7i–m** provided the product with the highest ee at 24 °C. Furthermore, alongside the slower reaction rates, the ee values of (*R*)-**15** also decreased at lower temperatures. Table 1 lists the ee values measured under the above-mentioned conditions.

Table 1 Results of Addition of Diethylzinc to Benzaldehyde in the Presence of Ligands (*S*)-**6** and (*R*)-**7**[a]

Entry	Ligand (*S*)-**6**	ee (*S*)-**15** (%)[b,c]	Ligand (*R*)-**7**	ee (*R*)-**15** (%)[b,d]
1	**6a**	86	**7a**	82
2	**6b**	96	**7b**	9
3	**6c**	92	**7c**	2
4	**6d**	94	**7d**	70
5	**6e**	92	–	–
6	**6f**	26	–	–
7	**6g**	90	–	–
8	**6h**	87	–	–
9	–	–	**7i**	90
10	–	–	**7j**	86
11	–	–	**7k**	18
12	–	–	**7l**	94
13	–	–	**7m**	91

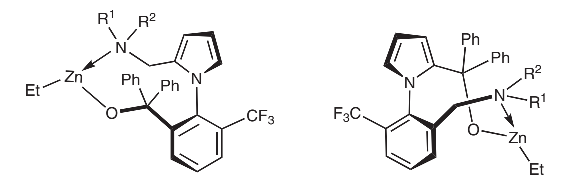
[a] The reactions were carried out with a 1 M solution of Et_2Zn in hexane (3 equiv).
[b] Determined by GC with a β-DEX™ 120 capillary column.
[c] Reactions carried out at 0 °C.
[d] Reactions carried out at 24 °C.

The steric bulk of the N-substituents significantly influenced the asymmetric induction. In the (*S*)-**6** series, the best results were achieved with **6b**, which contained an *N*,*N*-diethylamino group (96%). Smaller (**6a**) or larger (**6c**) substituents decreased the ee of the product (86 and 92%, respectively), whereas the chiral-induction efficiency of the pyrrolidin-1-yl-group-containing ligand **6d** was intermediate between those of **6b** and **6c** (94%). The presence of an additional chirality center in **6e** and **6f** had either no effect (**6e**) or a negative effect (**6f**) on the ee value of the product (92 and 26%, respectively), depending on the configuration of the asymmetric carbon atom in the ligand. However, the configuration of the major enantiomer was the same in both cases. This observation confirmed the crucial role of axial chirality in the asymmetric-induction process.

In case of ligands **7a–d**, the smallest N-substituent-containing ligand **7a** gave the best result (82% ee) but its asymmetric induction effect was significantly smaller than that of **6b**, the best ligand in the regioisomeric series. The ee of the product dropped to 9% and 2% in the presence of ligands containing bulkier *N*-substituents (**7b** and **7c**, respectively). However, the pyrrolidine moiety-containing **7d** gave a much better result (70%), presumably due to its more planar structure compared with the diethylamino and dibutylamino substituents. Similar observations have been reported in the case of stereogenic-center-containing ligands by Pericàs and co-workers,[31] but an opposite tendency was observed by Bringmann and Breuning[23] in the case of atropisomeric biaryl-type ligands. In that case, the best results were observed with a dibutylamino-group-containing ligand; however, the compound contained a phenolic OH group, which is much smaller than the tertiary alcohol functions of our ligands.

On the basis of reports in the literature[4,32] and our present results, one can conclude that, as well as the steric effects of the N-substituents, the electronic properties of the *tert*-amino and *tert*-hydroxy groups also have important roles in the formation of a zinc-containing catalyst in situ and in the subsequent enantioselective addition reaction in the presence of the regioisomeric ligand homologues (**6a–d** versus **7a–d**). In fact, in ligands **6a–d** the (dialkylamino)methyl group is connected to the electron-rich pyrrole ring, whereas the diaryl(hydroxy)methyl group is attached to the benzene ring possessing the electron-withdrawing trifluoromethyl group. In series **7a–d**, the reverse situation is observed, wherein the electron-donating ability of the amino nitrogen atoms and the acidity of the hydroxy groups are different in the regioisomeric pairs. Proposed structures of the catalysts **16** and **17** formed in situ are shown in Figure 2.

We surmised that the acidity of the zinc alcoholate-forming OH group might be increased in catalysts of series **17** if electron-withdrawing substituents were introduced into the neighboring phenyl groups. Therefore, ligands **7i–m** were also prepared and, for comparison, the same modification was carried out on the best ligand of the regioisomeric series, providing amino alcohols **6g** and **6h**. The results of the enantioselective diethylzinc addition to benzaldehyde in the presence of these multiple-trifluoromethylphenyl group-containing ligands are also listed in Table 1.



16 [(*S*)-**6** type catalyst] **17** [(*R*)-**7** type catalyst]

Figure 2 Presumed structures of catalysts formed in the presence of the (*S*)-**6** and (*R*)-**7** ligands

Increasing the OH acidity in **6g** and **6h** significantly decreased their asymmetric-induction efficiencies (90% and 87%, respectively) compared with the parent ligand **6b** (Table 1, entry 2 compared with Table 1, entries 7 and 8), indicating that **6b** has the optimal acidity of the OH group. On the other hand, the introduction of electron-withdrawing groups onto the phenyl substituents of **7a** or **7d** resulted in significantly better asymmetric induction (reactions with ligands **7j–m**; Table 1, entries 9–13). The best ee (94%) was obtained in the presence of **7l**, in which both aryl substituents possessed two trifluoromethyl groups. Reasons for this improved efficiencies include the increased stability of the zinc–amino alcohol complex and the steric effects of the trifluoromethyl groups.

To develop a theoretical rationale for these experimental results, we carried out quantum-chemical calculations out by using the *Gaussian 09* program package.[33] The B3LYP-631g(d,p) method and basis set was used for the H, C, N, O, and F atoms, and the B3LYP-SDD-MDF10 set was used for the Zn atom. The effect of hexane as solvent was also considered during the calculations of rotation and bond energies. Relative bond energies were calculated by using isodesmic reactions (see Supporting Information, Figure S1).

According to the reaction mechanism postulated by Yamakawa and Noyori,[4] the catalyst formed in situ has to react with a molecule of diethylzinc and a molecule of benzaldehyde to form a ternary molecular complex. This then passes into a transition-state complex in which the new C–C bond can be formed during its transformation into a product–catalyst complex. It is known that the energy difference between the two diastereoisomeric transition states of the rate-determining step influences the enantiomeric excess of the product. However, valuable information can also be collected from investigations into and comparisons of the ground-state geometries of catalyst series **16** and **17**. Therefore bond energies of the OZn…N bonds as well as the OZn…N and Zn…O bond distances of catalyst of type **16** and **17** catalysts were calculated (Table 2).

Table 2 Calculated Gibbs Free Energy Values for the OZn...N bond, and the Calculated OZn...N and Zn...O Distances in Catalysts **16** and **17**

Catalyst	(*S*)- or (*R*)-**15** *ee* (%)	$\Delta G_{OZn...N}$ (kJ/mol)	$l_{OZn...N}$ (Å)	$l_{Zn...O}$ (Å)
16a	86	65.6	2.345	1.852
16b	96	53.6	2.324	1.858
16c	92	41.5	2.313	1.860
16d	94	63.6	2.345	1.857
16e	92	30.5	2.372	1.851
16f	26	71.6	2.309	1.893
16g	90	58.8	2.298	1.869
16h	87	72.0	2.278	1.879
17a	82	55.2	2.399	1.852
17b	9	31.6	2.426	1.849
17c	2	14.3	2.428	1.849
17d	70	81.0	2.401	1.856
17i	90	56.8	2.371	1.863
17j	83	77.0	2.372	1.864
17l	94	66.8	2.338	1.869

The calculated data suggest that both the electronic and steric properties of the N-substituents can influence the chiral-discriminating properties of catalysts of type **16**. Within the homologous series **16a**, **16b**, and **16c**, the OZn…N bond length decreases gradually in line with the increasing size of the N-substituent (Me, Et, Bu). However, the OZn…N bond energy also decreases due to the increasing bulk and decreasing electron-donating effect of the N-substituent. On the other hand, the O…Zn distance becomes longer within the catalyst complex. Consequently, smaller or larger substituents on the nitrogen atom strongly influence the accessibility of the zinc atom for further complex formation during the reaction. An optimal combination of these parameters can be found at catalyst **16b**, resulting in the highest ee. In **16d**, the pyrrolidine ring space requirements are smaller than the two butyl groups in **16c**, the OZn…N bond is more stable, and the OZn…N and Zn…O distances are between those for catalysts **16a** and **16b**. The same parameters of the asymmetric centers in **16e** and **16f** are quite different. In **16e**, the OZn…N bond energy is small, and relatively long OZn…N and short O…Zn distances can be found, so the observed high ee in the presence of this catalyst can be rationalized by taking into consideration the positive discriminative effect of the configuration in the asymmetric center containing the N-substituent. In **16f**, the OZn…N bond is very short, whereas the O…Zn distance is large, so it is probable that the configuration of the additional asymmetric center does not reinforce the axial-chirality-determined enantiopreference. The introduction of trifluoromethyl groups caused a significant modification of the structure of the catalyst. The electron-donating ability of the oxygen atoms in **16g** and **16h** is diminished compared with that of the parent **16b**; consequently, the O…Zn bonds are longer and the OZn…N bonds are shorter and stronger than the optimal values found in **16b**.

The regioisomeric ligand-containing catalysts of type **17** are more crowded around the amino nitrogen atom because of the different bond and torsion angles compared with those of type **16** catalysts. Therefore, within the series **17a–c**, the OZn…N bond length and bond energy are closest to the corresponding values for **16b** in the smallest dimethylamino-group-containing catalyst **16a**. Again, parameters for the pyrrolidino-group-containing catalyst **17d** are between those of **17a** and **17b**, so this catalyst shows good asymmetric induction. The calculated parameters of catalysts **7i–m** are in good accordance with the experimental findings. The electron-withdrawing trifluoromethyl groups overcompensate for the electron-donating effect of the pyrrole ring and increase the OH acidity in these catalyst ligands. Consequently, the Zn…O bond lengths are longer and the OZn…N bonds are shorter and stronger than those in **17a**. The investigated bond lengths in **17l** are the closest to those of the regioisomeric **16b**, and this catalyst produced optically active **15** with the highest ee.

On the basis of our experimental findings and quantum-chemical calculations, we conclude that interchanging the functional groups positions in the atropisomeric amino alcohols under investigation resulted in regioisomeric ligands in which both the electronic and the steric properties of the functional groups played a role. Systematic modifications of the neighboring substituents led us to optimize the corresponding properties of the amino and the hydroxy functions, providing excellent atropisomeric amino-alcohol-type ligands for enantioselective diethylzinc addition to aromatic aldehydes. Quantum-chemical calculations were in good accordance with our experimental results. This first systematic comparative study of regioisomeric series of atropisomeric ligands demonstrated that interchange of the functional-group positions can cause significant differences in the efficiency of the regioisomeric catalyst ligands as a result of changes in the electronic properties and steric accessibilities of the functional groups.

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Supporting Information

Supporting information for this article is available online at https://doi.org/10.1055/s-0037-1610551.

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(30) **Addition of Diethylzinc to Benzaldehyde: General Procedure**
The appropriate ligand (*S*)-**6** or (*R*)-**7** (0.095 mmol, 99% ee) was dissolved in a 1 M solution of Et_2Zn in hexane (0.6 mL, 0.6 mmol) under N_2, and the mixture was stirred for 1 h at r.t. while the temperature was adjusted to 24 °C or 0 °C. Freshly distilled PhCHO (0.2 mmol) was added and the color of the mixture changed to a distinctive yellow. After stirring for 16 h, the mixture again became colorless, indicating completion of the reaction. The reaction was quenched by addition of sat. aq NH_4Cl (5 mL), and the mixture was extracted with toluene (3 × 5 mL). The organic layers were combined, dried (Na_2SO_4), filtered, and concentrated under reduced pressure. Purification of the residue by column chromatography (silica gel, EtOAc–hexane) gave 1-phenylpropan-1-ol (**15**) as a colourless oil. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.26 (m, 5 H), 4.60 (t, *J* = 6.5 Hz, 1 H), 1.93–1.64 (m, 3 H), 0.92 (t, *J* = 7.4 Hz, 3 H). The ee was determined by GC analysis: Supelco-DEX™ 120 capillary column (0.25 nm/0.25 μm, 30 m), T_{Inj}: 250 °C, T_{Det}: 250 °C (FID), N_2: 1 mL/min, split: 100:1, oven: 60 °C → 140 °C (10 °C/min). R_T: (*R*)-enantiomer 18.3 min; (*S*)-enantiomer 18.6 min.
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